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Please find below the README information of the required contents for code and software submission checklist. We provide step-by-step guidance to run the code to fully replicate the main findings of our work entitled "**Neural dynamics of social impression updating during narrative comprehension**". These instructions include downloading the preprocessed fMRI data, running the key analyses, and plotting figures for publication.

1. System requirements

Please set up the virtual environment, socialaha, with 'conda env create -f environment.yml' and 'conda activate' it when you run the python scripts. In addition, step03 and step05 require RStudio Version 2024.04.2+764;

2. Installation guide

RStudio: https://posit.co/downloads/
Visual Studio code: https://code.visualstudio.com/
Anaconda: https://www.anaconda.com/download
Jupyter: https://jupyter.org/install

3. Demo

Complied source code could be downloaded here: https://github.com/jinke828/socialaha/code
Behavioral data could be found here: https://github.com/jinke828/socialaha/data/beh
fMRI data could be downloaded from OpenNeuro: https://openneuro.org/datasets/ds005658.

Upon downloading the required software, data and code, we suggest that you run the scripts by this following order. Note that most of these scripts have set the default dictionary, for example, `os.chdir(' /gpfs/milgram/project/chun/jk2992/socialaha/')`. Please modify it to the path of your downloaded 'socialaha' folder.

step01_load-brain.ipynb

Loads raw preprocessed fMRI NIfTI files for all participants and extracts time series for the 116 ROIs (Schaefer 100-parcel cortical atlas and Tian 16-region subcortical atlas). Censores motion-artifact TRs, z-scores voxel time series, and applies a 4TR hemodynamic correction. Segments the BOLD signal into movie-watching events and impression-rating periods using onset/offset times from event `.tsv` files. This step will take 45 to 90min depending on the computing power of your device

step02_count_aha.ipynb

Quantifies aha moment frequency across subjects by counting screenshot image files in each participant's `ahascreenshots/` block directories. Generates descriptive statistics (mean, max, SD) and a histogram of the aha count distribution. This step takes less than a minute.

step03_impression-updates.py + step03_plot_impression_updates.R
Analyzes how participants' verbal impressions of story characters evolve across the 10 viewing sessions. Embeds speech transcriptions using Google Universal Sentence Encoder (512-dim vectors) and measures cosine similarity between impressions. Fits a nested linear mixed-effects model (`similarity ~ distance + (1 | group/sub) + (1 | character)`) to the impression similarity data from step03, then plots mean similarity by temporal distance with error bars. This step takes less than 5 minutes.

step04_IS-RSA.py
Tests whether neural inter-subject correlation (ISC) during movie watching predicts between-subject similarity in post-movie character impressions (IS-RSA). Computes Spearman correlation between Fisher z-transformed neural ISC and impression similarity across 116 ROIs, with 10,000 bootstrap iterations and FDR correction. Runs two analyses: after-movie ISC (all 10 runs) and before-movie ISC (brain runs 2–10 vs. impressions runs 1–9, 1-run lag). Generates surface-mapped brain visualizations on Fsaverage5. This step takes 10-20 minutes.

step05a_IS-RSA-mediation.ipynb + step05b_mediation.R
Prepares the data matrix for mediation analysis by assembling pre-impression similarity (runs 1–9, i.e., impression before each movie), post-impression similarity (runs 2–10, i.e., impression after each movie), and neural ISC values for all 116 ROIs into a single long-form dataframe indexed by scene and subject pair. This step takes less than a minute.

Performs caudal mediation analysis for all 116 ROIs testing whether neural ISC mediates the relationship between pre- and post-movie impression similarity. For each ROI, fits two lm models (mediator and outcome, controlling for scene, pair, condition) and runs `mediate()` with 1,000 bootstrap iterations. Extracts ACME (indirect effect), ADE (direct effect), total effect, and proportion mediated. This step takes 1-2 hours.

step05c_mediation_rSTS.R
Focused replication of the mediation analysis for ROI 98 (right Superior Temporal Sulcus) alone, using 10,000 bootstrap iterations for greater precision. Also generates the full null ACME bootstrap distribution for visualization. This step takes 10-30 minutes.

step06_compute_mtm_neural-pattern-shift.ipynb
Computes moment-to-moment neural pattern dissimilarity (1 − Pearson r between consecutive TRs) for each participant, run, and ROI. Iterates over 3 groups × 116 ROIs × subjects × runs and stores the resulting arrays. This step takes 1-2 hours.

step07_neural-pattern-shift_social-insight.py

Tests whether neural pattern shifts are elevated around character-related aha moments relative to chance. Filters aha annotations to character-related events (multi-rater consensus ≥ 2 of 3), extracts pattern shift values from -5 to $+3$ TRs around each aha, and compares to a null distribution of 10,000 random non-aha timepoints per ROI. Applies two-stage FDR correction across ROIs and time points. Identifies ROIs with significant shifts and visualizes z-transformed shifts on brain surface maps. This step takes 10-20 minutes.

step08_neural-pattern-shift-impression_update.py

Tests whether neural pattern shifts around character aha moments correlate with impression updates across subjects. Computes impression update as $1 -$ cosine similarity between consecutive run embeddings per character. For each ROI $\times$ TR, computes scene-wise Spearman r between pattern shifts (Fisher z-transformed) and impression updates, then averages across 40 scenes. Null distribution uses 1,000 permutations. Applies two thresholds: step08-only FDR (cortical ROIs) and a double-threshold requiring overlap with step07-significant ROIs. Generates brain surface maps per TR. This step takes 5-10 minutes.

Please do not hesitate to reach out to Jin or Hayoung with questions.

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On behalf of all authors